

CarelioEMR — Comprehensive Technical Documentation

****Subject:**** Site Admin Configuration & Subscription Management Modules

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1. Executive Summary & Architecture Overview

CarelioEMR is an enterprise multi-tenant Electronic Medical Records (EMR) platform based on OpenEMR. To ensure long-term maintainability, seamless upstream OpenEMR upgrades, and tenant isolation, all platform extensions are built strictly according to ****Zero Core Modification Principles****.

Key Architectural Guidelines:

1. ****Zero Core Modifications:**** No core OpenEMR engine scripts (in ``/src/``, ``/interface/main/``, ``/library/``) are altered.

2. ****Custom Module Encapsulation:**** All capabilities are bundled as independent, self-contained OpenEMR modules in ``/interface/modules/custom_modules/``:

****`site\_admin\_config`:**** Tenant Administration, Caribbean Demographics, Branding Assets, Role-Based Access Control, and Temporary Password Security.

****`carelio\_subscription`:**** Subscription Lifecycle Monitoring, 2-Day Grace Period, Safe Deactivation, Site Admin Renewals/Extensions, Server-Side Access Enforcement, and Daily Automated Background Monitoring.

3. ****Idempotency & Zero Data Loss:**** All database scripts and migrations execute with ``ON DUPLICATE KEY UPDATE`` and idempotent checks. Subscription deactivation never deletes clinical, patient, encounters, or billing data.

2. Module 1: Site Admin Configuration (`site\_admin\_config`)

2.1 Caribbean Geographic Demographics Engine

****Dataset Scope:**** Embedded full geographic coverage for 29 Caribbean countries, all 10 Saint Lucia districts, and 367 distinct local communities.

****Storage & Structure:**** Self-contained within `table.sql` and `sql/table.sql` (90 KB+). Data is injected into OpenEMR's standard `list\_options` schema using composite primary keys `(list\_id, option\_id)` with `ON DUPLICATE KEY UPDATE`.

****Dynamic Cascading UI Script:**** Embedded `location\_cascading\_script` into `layout\_options`. When a user selects a Caribbean country (e.g., Saint Lucia `LC`), the state dropdown dynamically repopulates with the 10 administrative districts. Selecting a district filters communities accordingly.

2.2 Carelio Brand Assets Auto-Deployment

Implemented in `SiteAdminInstaller::deployBrandAssets()`.

Automatically provisions Carelio branding assets to tenant site directories (`oemr/sites/<tenant>/images/logos/`):

`core/login/primary` & `portal/login/primary` → `logo.svg`, `logo.png`

`core/favicon` → `favicon.ico`

`core/menu/primary` & `portal/menu/primary` → `logo.svg`, `logo.png`

Legacy fallbacks → `login\_logo.gif`, `logo\_1.png`, `logo\_2.png`

2.3 Robust Multi-Runtime Database Connection Resolver

****Issue Solved:**** In OpenEMR web runtime, `\$GLOBALS['adodb']['db']->\_connectionID` is a `mysqli` instance, while CLI/seeder use `PDO`.

****Implementation (`SiteAdminInstaller::resolvePdo()`):****

1. Checks for existing PDO instances in `\$GLOBALS['dbh']`.

2. Resolves target tenant site directory via `\$GLOBALS['OE\_SITE\_DIR']`, session, or folder inspection.

3. Uses OpenEMR `DatabaseConnectionOptions::forSite()` and `DatabaseConnectionFactory::createDbal()`.

4. Parses tenant `sqlconf.php` in an isolated scope and establishes a dedicated UTF-8 PDO connection (`SET NAMES 'utf8mb4', sql\_mode = "`).

2.4 Forced Temporary Password Security

****Files:**** `TemporaryPasswordService.php` and `change\_temporary\_password.php`.

****Flow:**** First-time tenant login triggers mandatory password reset. Validates current temporary password, verifies complexity, updates hash via OpenEMR `AuthUtils`, clears temporary flag, and immediately redirects to the OpenEMR login page (`login.php?site=<tenant>`) for a clean, secure session launch.

3. Module 2: Subscription Management (`carelio\_subscription`)

3.1 Status Lifecycle State Machine

The subscription module enforces a 4-stage lifecycle state machine:



Lifecycle Stage Details:

1. **ACTIVE**:

Subscription is active. Account has full access.

2. **EXPIRING_SOON**:

Remaining days ≤ 2 days (48 hours).

Daily monitoring process flags subscription and dispatches renewal reminder notification to clinic administrator.

3. **EXPIRED_GRACE_PERIOD**:

Period end has passed, but time is within the 2-day (48-hour) grace window.

Clinic users can continue using the application without interruption.

Prominent grace period notice displayed on the dashboard.

4. ****DEACTIVATED****

2-day grace period has elapsed without renewal.

Site is deactivated and locked server-side.

****Data Preservation Guarantee:**** Zero data deletion. All clinical records, encounters, medical history, documents, and billing records remain 100% intact.

Account remains immediately available for Site Administrator reactivation.

3.2 Database Schema Architecture

The module utilizes 4 dedicated database tables inside each tenant database:

```
-- 1. Main Subscriptions Table
CREATE TABLE IF NOT EXISTS `mod_carelio_subscriptions` (
  `id` INT AUTO_INCREMENT PRIMARY KEY,
  `external_subscription_id` VARCHAR(191) NULL UNIQUE,
  `site_id` VARCHAR(191) NULL,
  `site_name` VARCHAR(191) NULL,
  `tenant_slug` VARCHAR(191) NULL,
  `customer_name` VARCHAR(191) NULL,
  `customer_email` VARCHAR(191) NULL,
  `plan_name` VARCHAR(120) NULL,
  `plan_amount` DECIMAL(10,2) NOT NULL DEFAULT 0.00,
  `currency` VARCHAR(10) NOT NULL DEFAULT 'USD',
  `billing_cycle` VARCHAR(20) NOT NULL DEFAULT 'monthly',
  `status` VARCHAR(40) NOT NULL DEFAULT 'ACTIVE',
  `current_period_start` DATETIME NULL,
  `current_period_end` DATETIME NULL,
  `expiration_reminder_sent_at` DATETIME NULL,
  `grace_started_at` DATETIME NULL,
  `grace_ends_at` DATETIME NULL,
  `deactivated_at` DATETIME NULL,
  `renewed_at` DATETIME NULL,
  `reactivated_at` DATETIME NULL,
  `created_at` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  `updated_at` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  KEY `idx_carelio_subscription_site` (`site_id`),
  KEY `idx_carelio_subscription_status` (`status`),
  KEY `idx_carelio_subscription_period_end` (`current_period_end`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 2. Configuration Settings Table
CREATE TABLE IF NOT EXISTS `mod_carelio_subscription_config` (
  `id` INT AUTO_INCREMENT PRIMARY KEY,
  `config_key` VARCHAR(120) NOT NULL UNIQUE,
  `config_value` TEXT NULL,
  `updated_at` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
```

```

) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 3. Audit History & Event Log
CREATE TABLE IF NOT EXISTS `mod_carelio_subscription_history` (
  `id` BIGINT AUTO_INCREMENT PRIMARY KEY,
  `subscription_id` INT NULL,
  `action` VARCHAR(120) NOT NULL,
  `old_status` VARCHAR(40) NULL,
  `new_status` VARCHAR(40) NULL,
  `old_period_end` DATETIME NULL,
  `new_period_end` DATETIME NULL,
  `performed_by` VARCHAR(191) NULL,
  `notes` TEXT NULL,
  `created_at` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  KEY `idx_carelio_sub_hist_sub` (`subscription_id`),
  KEY `idx_carelio_sub_hist_act` (`action`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 4. External Webhook Events Table
CREATE TABLE IF NOT EXISTS `mod_carelio_subscription_events` (
  `id` BIGINT AUTO_INCREMENT PRIMARY KEY,
  `subscription_id` INT NULL,
  `event_type` VARCHAR(120) NOT NULL,
  `event_payload` LONGTEXT NULL,
  `created_at` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

3.3 Site Administrator Dashboard & Actions (`public/index.php`)

****Navigation Path:**** OpenEMR Navigation Menu → ****Administration → Carelio Subscription****

****Access Control:**** Restricted via OpenEMR Core ACL (`admin`, `users`). Regular clinicians and staff receive a read-only notification without modification controls.

****Key Components:****

1. ****Status & Expiration Banner:**** Real-time color-coded alerts (Green: Active, Yellow: Expiring Soon, Orange: Grace Period, Red: Deactivated).

2. ****Plan & Billing Metrics:**** Plan Name, Monthly Fee, Start Date, Expiry Date, and Grace Window Countdown.

3. ****Site Admin Action Controls:****

****Renew Subscription (+1 Month):**** Automatically calculates future date (+30 days/month), restores status to `ACTIVE`, and records audit history.

****Extend Expiration Date:**** Interactive date picker allowing custom extension dates.

****Reactivate Subscription:**** One-click instant reactivation for sites in `DEACTIVATED` state.

4. ****Interactive Simulator Toolbar:****

Dedicated testing toolbar allowing instant switching between all 4 statuses (`ACTIVE`, `EXPIRING\_SOON`, `EXPIRED\_GRACE\_PERIOD`, `DEACTIVATED`) without waiting 30 days.

5. ****Subscription Audit History Table:****

Comprehensive immutable audit log tracking every action, status transition, timestamp, user, and notes.

3.4 Server-Side Access Enforcement (`SubscriptionEnforcementService.php`)

To prevent users from bypassing restrictions through direct URL bookmarks or REST API calls:

Registered inside `openemr.bootstrap.php` on every authenticated web request.

Behavior when Status is `DEACTIVATED`:

Site Administrator: Automatically redirected to `carelio\_subscription/public/index.php` with reactivation controls enabled.

Regular Staff / Clinicians: HTTP 403 Forbidden page rendered with the Carelio "Account Deactivated" security template explaining that data is preserved and advising them to contact the Site Administrator.

Login, logout, assets, and subscription renewal paths are exempt from interception.

3.5 Automated Daily Monitoring Engine (`CarelioCheckSubscriptionsCommand.php`)

Command: `php artisan carelio:check-subscriptions`

Execution Flow:

1. Scans all tenant databases across MySQL.
 2. Evaluates expiration timestamps against current system time.
 3. Detects subscriptions expiring within 2 days $\rightarrow$ updates status to `EXPIRING\_SOON` and triggers renewal reminders.
 4. Detects expired subscriptions within 2-day window $\rightarrow$ updates status to `EXPIRED\_GRACE\_PERIOD`.
 5. Detects subscriptions expired past 2-day grace $\rightarrow$ sets status to `DEACTIVATED` and logs deactivation event.
 6. Renders live execution summary table in console/logs.
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4. Live Server Deployment & Verification Guide

Step 1: Push Local Branch & Pull on Live Server

On development workstation:

```
git add .
git commit -m "feat: complete subscription management module with monitoring, grace period, renewal, and background cron"
git push origin subscription-model
```

On live server terminal:

```
cd /path/to/carelioemr
git fetch origin
git checkout subscription-model
git pull origin subscription-model
```

Step 2: Module Activation in OpenEMR UI

1. Sign in to OpenEMR as Super Administrator (`admin` / `pass`).
2. Navigate to: **Administration → System → Modules**.
3. Under the **Unregistered** tab: locate **Carelio Subscription** and click **Register**.
4. Under the **Registered** tab: click **Install SQL**, then click **Enable**.

(Note: For newly provisioned tenants, `carelio\_subscription` is pre-registered in the tenant baseline SQL).

Step 3: Configure Live Background Cron Job

Add the daily subscription monitoring process to the live server crontab:

```
crontab -e
```

Add the following entry:

```
0 0 * * * cd /path/to/carelioemr && php artisan carelio:check-subscriptions >> /var/log/carelio_sub_cron.log 2>&1
```

Step 4: Live Verification Checklist

Item	Test Action	Expected Result	Verified
1	Log in as `siteadmin` on live tenant	Menu contains Administration → Carelio Subscription	[]
2	Open Subscription Dashboard	Displays Plan, Status Badge (`ACTIVE`), Expiry Date	[]

3	Click ***"Test EXPIRING SOON"***	Banner turns Yellow, status becomes `EXPIRING_SOON`	[]
4	Click ***"Test GRACE PERIOD"***	Banner turns Orange, grace window active	[]
5	Click ***"Test DEACTIVATED"***	Banner turns Red, server-side guard engages	[]
6	Click ***"Renew Subscription (+ 1 Month)"***	Status returns to `ACTIVE`, date advances +30 days	[]
7	Check Audit History Table	All test events logged with timestamps and actor	[]
8	Run CLI command	`php artisan carelio:check-subscriptions` outputs tenant table	[]

5. Summary of Files Changed & Created

File Path	Purpose
`oemr/interface/modules/custom_modules/carelio_subscription/ModuleManagerListener.php`	OpenEMR UI module lifecycle dispatcher (install, enable, upgrade, disable).
`oemr/interface/modules/custom_modules/carelio_subscription/openemr.bootstrap.php`	PSR-4 module autoloader, menu subscriber, and enforcement guard hook.
`oemr/interface/modules/custom_modules/carelio_subscription/public/index.php`	Full Site Administrator Subscription Dashboard, actions, simulator, and audit log UI.
`oemr/interface/modules/custom_modules/carelio_subscription/src/Services/SubscriptionManagerService.php`	Core business logic for renewals, extensions, status synchronization, and simulation.
`oemr/interface/modules/custom_modules/carelio_subscription/src/Security/SubscriptionEnforcementService.php`	Server-side request interceptor blocking access during deactivated status.
`oemr/interface/modules/custom_modules/carelio_subscription/src/Installer/SubscriptionInstaller.php`	Robust multi-context database installer and schema manager.
`oemr/interface/modules/custom_modules/carelio_subscription/table.sql`	Self-contained schema definition for

	the 4 subscription tables and defaults.
<code>`app/Console/Commands/CarelioCheckSubscriptionsCommand.php`</code>	Automated daily artisan command to check expirations, grace periods, and deactivations.
<code>`oemr/interface/modules/custom_modules/site_admin_config/public/change_temporary_password.php`</code>	Secure temporary password update with clean login redirection.
<code>`oemr/interface/modules/custom_modules/site_admin_config/ModuleManagerListener.php`</code>	Added upgrade and upgrade_sql lifecycle hooks.